Problem

Problem · Triangle inequality and the case of equality.

1. Let z be a complex number. Prove that $\mathrm{Re}(z) \leq |z|$, with equality if and only if z is a nonnegative real number.
2. Let z and z' be two complex numbers. Prove that $|z+z'|^2 = |z|^2 + |z'|^2 + 2\,\mathrm{Re}\left(z\overline{z'}\right)$.
3. Deduce the triangle inequality $|z+z'| \leq |z| + |z'|$.
4. Assume $z' \neq 0$. Prove that equality $|z+z'| = |z| + |z'|$ holds if and only if there is a real number $\lambda \geq 0$ such that $z = \lambda z'$. Interpret with the images M and M' of z and z'.
5. Application. Consider the points M with affix z such that $|z-3-4i| \leq 2$.
   (a) Prove that for these points, $3 \leq |z| \leq 7$.
   (b) Determine the affixes of the points for which $|z| = 7$, then $|z| = 3$.